GUIDELINES 2025
EUROPEAN RESUSCITATION COUNCIL©

ADULT PERI-ARREST ALGORITHM*

Assess with ABCDE approach

- Monitor SpO_2, ECG, BP
- Give oxygen if SpO_2 < 94% and obtain IV access
- Record 12-lead ECG
- Identify and treat reversible causes

Life-threatening features?

- Shock
- Syncope with severe or ongoing hypotension
- Myocardial ischaemia
- Severe heart failure with pulmonary oedema or shock
- Or immediately after ROSC

YES

UNSTABLE

Synchronised shock

- Up to 3 attempts
- Sedation / anaesthesia if conscious

If unsuccessful

- Amiodarone 300 mg IV over 10-20 min, or Procainamide 10-15 mg kg^{-1} (max 1g) over 20 min
- Repeat synchronised shock

NO

This part of the algorithm does not address every possible scenario

STABLE

Evaluate QRS

Broad (≥ 0.12 s)

Irregular

Possible actions include:

Atrial fibrillation with pre-excitation:
Procainamide or cardioversion

Polymorphic VT with QT prolongation:

- Give Mg^{2+} 8 mmol IV over 10 min
- Consider Isoprenaline or temporary pacing to increase HR
- Avoid Amiodarone

Regular

Known VT, vagal manoeuvres and adenosine fail, uncertain mechanism of arrhythmia
or known or suspected heart disease

If high sedation / anaesthesia risk, give Procainamide 10-15 mg kg^{-1} over 20 min

If Procainamide unavailable or contraindicated (e.g. severe HF, acute MI or end-stage renal disease), give Amiodarone 300 mg IV over 10-60 min, then 900 mg over 24 h

Narrow (< 0.12 s)

Regular

- **Vagal manoeuvres**
- If ineffective: **Adenosine** (if no preexcitation) 6 mg rapid IV bolus, if unsuccessful give 12 mg, if unsuccessful give 18 mg
- If ineffective: Verapamil or Beta-blocker
- If ineffective: **synchronised shock**

Irregular

Probable atrial fibrillation:

- EF > 40%: Control rate with a Beta-blocker, Verapamil, Diltiazem, or Digoxin
- EF < 40%: Consider Beta-blocker or Digoxin
- Anticoagulate if duration of the arrhythmia > 24 h

Synchronised shock

- Sedation / anaesthesia if conscious

Seek expert help

*This algorithm is for peri-arrest tachyarrhythmia of abnormal origin. It is not for sinus tachycardia where the underlying cause should be treated - do not try to normalise the heart rate in sinus tachycardia with antiarrhythmic medication or synchronised cardioversion.

Seek expert help if the tachyarrhythmia and, or life-threatening features persist.

SCAN FOR MORE HELPFUL MATERIALS

ADULT BRADYCARDIA ALGORITHM

ADVANCED LIFE SUPPORT

- ASSESS with ABCDE approach
- Monitor SpO_2, ECG, BP
- Give oxygen if SpO_2 < 94% and obtain IV access
- Record 12 lead ECG
- Identify and treat reversible causes (e.g. myocardial infarction or electrolyte abnormalities)

Life-threatening features?

1. Shock
2. Syncope
3. Myocardial ischaemia
4. Severe heart failure

Or immediately after ROSC

YES

Atropine 500 mcg IV

Satisfactory response?

NO

Consider interim measures:

- Atropine 500 mcg IV, repeat to maximum of 3 mg
- Isoprenaline 5 mcg min^{-1} IV
- Adrenaline 2-10 mcg min^{-1} IV
- Alternative drugs*

and / or

- Transcutaneous pacing

Seek expert help
Arrange transvenous pacing

YES

NO

Risk of asystole?

- Recent asystole
- Mobitz II atrio-ventricular block
- Complete heart block with broad QRS
- Ventricular pause > 3s

YES

NO

Observe

***Alternatives include:**

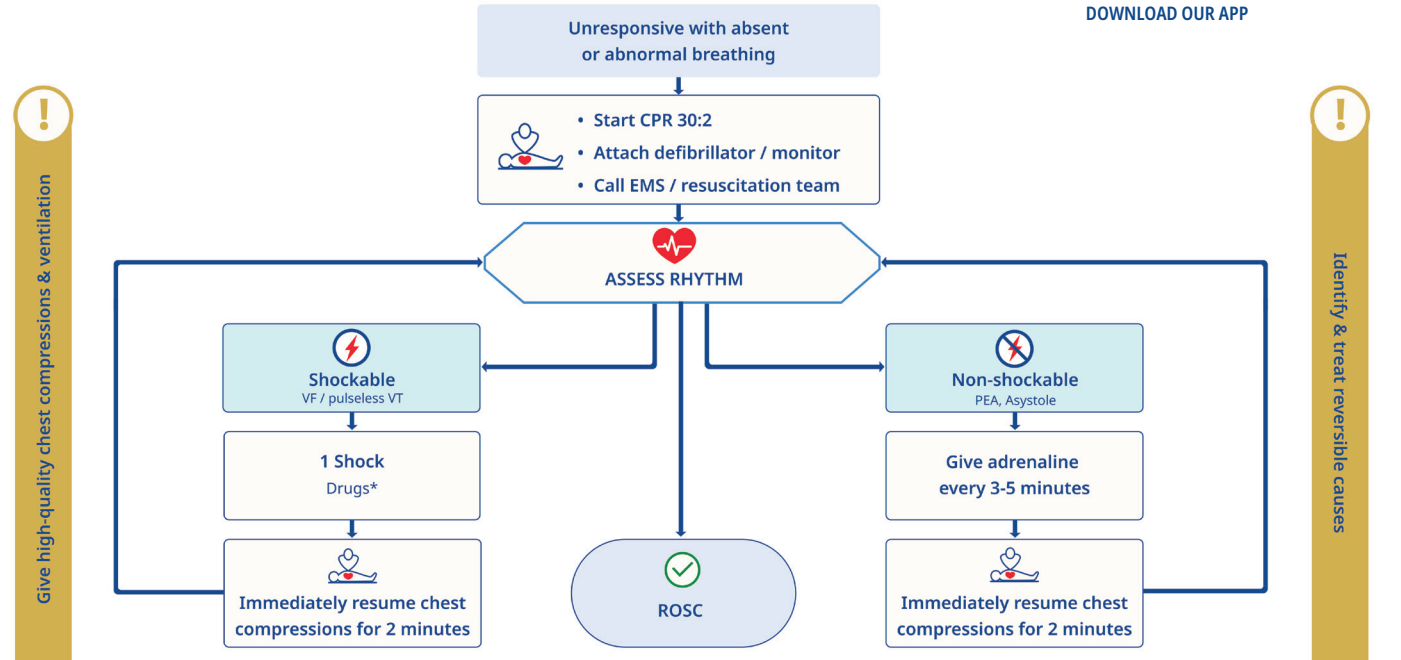
- Aminophylline
- Dopamine
- Glucagon (if bradycardia is caused by beta-blocker or calcium channel blocker)
- Glycopyrrolate (may be used instead of Atropine)

GUIDELINES 2025
EUROPEAN RESUSCITATION COUNCIL©

ADULT ADVANCED LIFE SUPPORT ALGORITHM

DOWNLOAD OUR APP

ADVANCED LIFE SUPPORT



- Ensure optimal defibrillator pad placement
- Give oxygen
- Continuous compressions if tracheal tube or supraglottic airway
- Use waveform capnography
- Minimise interruptions to chest compressions
- Early IV access (IO if IV not possible)

* Drugs

- Give 1st adrenaline after 3 shocks, then every 3-5 minutes
- Give amiodarone after 3 shocks

Consider

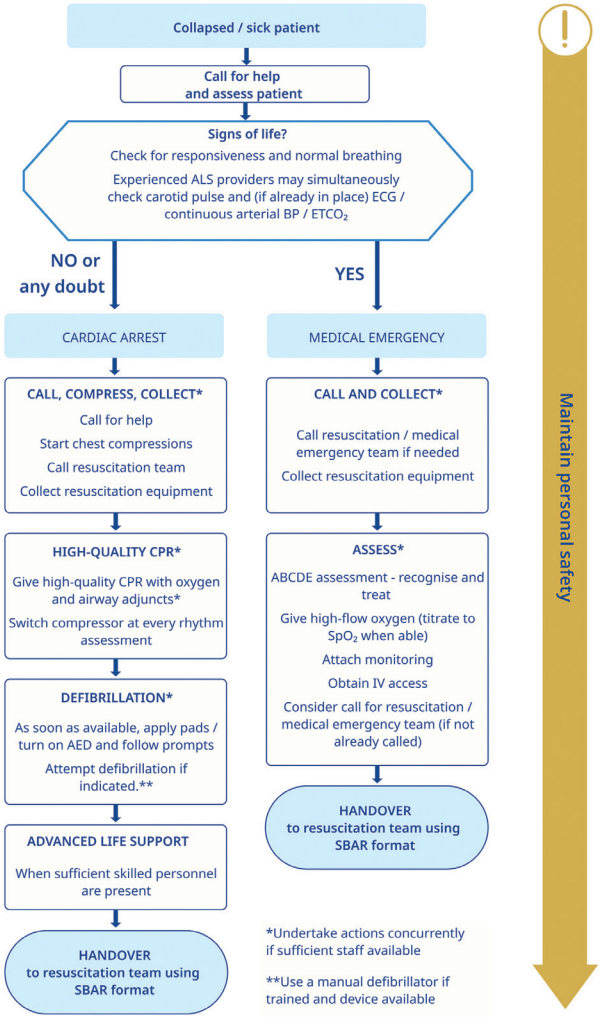
- Changing pads to antero-posterior after 3 shocks
- Mechanical chest compressions to facilitate transfer / treatment
- Ultrasound to identify reversible causes
- Extracorporeal CPR and, or Coronary angiography/percutaneous coronary intervention

Immediately after ROSC

- Use ABCDE approach
- Aim for SpO_2 of 94-98% and normal $PaCO_2$
- Aim SBP > 100 mmHg
- 12 Lead ECG
- Identify and treat cause
- Temperature control

- Hypoxia
- Hypovolaemia
- Hyper-hypokalaemia / metabolic
- Hypothermia, hyperthermia
- Toxins
- Tamponade (cardiac)
- Tension pneumothorax
- Thrombosis (coronary / pulmonary)

IN-HOSPITAL RESUSCITATION ALGORITHM



*Undertake actions concurrently if sufficient staff available

**Use a manual defibrillator if trained and device available